



Introduction to CFD (4RC30)

The science of scientific writing *or*
Writing for Readers

Prof.dr.ir. Niels Deen, N.G.Deen@tue.nl, Tel. 3681, VEC 3.202

Dr. Yali Tang, y.tang2@tue.nl, Tel. 8052, VEC 3.106

Scientists Unveil New 'Tree of Life'

By CARL ZIMMER APRIL 11, 2016

A team of scientists unveiled a new tree of life on Monday, a diagram outlining the evolution of all living things. The researchers found that bacteria make up most of life's branches. And they found that much of that diversity has been waiting in plain sight to be discovered, dwelling in river mud and meadow soils.

"It is a momentous discovery — an entire continent of life-forms," said Eugene V. Koonin of the National Center for Biotechnology Information, who was not involved in the study.

The [study was published](#) in the journal Nature Microbiology.

In his 1859 book "On the Origin of Species," Charles Darwin envisioned evolution like a branching tree. The "great Tree of Life," he said, "fills with its dead and broken branches the crust of the earth, and covers the surface with its ever branching and beautiful ramifications."

Ever since, biologists have sought to draw the tree of life. The invention of DNA sequencing revolutionized that project, because scientists could find the relationship among species encoded in their genes.

In the 1970s, Carl Woese of the University of Illinois and his colleagues published the first "universal tree of life" based on this approach. They presented the tree as three great trunks.

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Scientific writing

**If the reader is to grasp what the writer means, the writer must understand what
the reader needs
or**

Accurate information is useless if the reader cannot understand it

References:

Gopen and Swan (1990), The science of scientific writing, *American Scientist*, 78, 550

Hale (2005), www.ou.edu/englhale/gopenswanrev.ppt

Substance and structure

$t(\text{time})=15', T(\text{temperature})=32^\circ, t=0', T=25^\circ; t=6', T=29^\circ;$
 $t=3', T=27^\circ; t=12', T=32^\circ; t=9', T=31^\circ$

Substance and structure

Temperature(°C)	Time (min)
25	0
27	3
29	6
31	9
32	12
32	15

Substance and structure

Time (min)	Temperature(°C)
0	25
3	27
6	29
9	31
12	32
15	32

- Give context before data (“cause before effect”).
- The information which is familiar (time increments) comes before the “new” data.

Why is this text so hard to read?

The smallest of the URF's (URFA6L), a 207-nucleotide (nt) reading frame overlapping out of phase the NH₂-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene has been identified as the animal equivalent of the recently discovered yeast H⁺-ATPase subunit 8 gene. The functional significance of the other URF's has been, on the contrary, elusive. Recently, however, immunoprecipitation experiments with antibodies to purified, rotenone-sensitive NADH-ubiquinone oxido-reductase [hereafter referred to as respiratory chain NADH dehydrogenase or complex I] from bovine heart, as well as enzyme fractionation studies, have indicated that six human URF's (that is, URF1, URF2, URF3, URF4, URF4L, and URF5, hereafter referred to as ND1, ND2, ND3, ND4, ND4L, and ND5) encode subunits of complex I. This is a large complex that also contains many subunits synthesized in the cytoplasm.

What if we remove the difficult words?

The smallest of the URF's, and [A], has been identified as a [B] subunit 8 gene. The functional significance of the other URF's has been, on the contrary, elusive. Recently, however, [C] experiments, as well as [D] studies, have indicated that six human URF's [1-6] encode subunits of Complex I. This is a large complex that also contains many subunits synthesized in the cytoplasm.

Improving accessibility

- **Grammatical subjects should be followed as soon as possible by their verbs**
 - Avoid:
I would, because of many reasons (the first of which is location, the second of which is challenge, the third of which is independence) like to remain my own boss.
- **Every unit of your story should serve a *single function* or make a *single point***
 - Avoid:
I have received a Master's degree from the Eindhoven University of Technology and, since I live nearby, can come to an interview at any time.
- **Information that you want to emphasize should appear at points of syntactic closure**
 - Syntactic closure occurs at the end of a clause or sentence
Syntactic closure comes HERE, but it can also come HERE.

Subject-verb separation

- **Subject and verb are very much separated:**

The **smallest of the URF's** is URFA6L, a 207-nucleotide (nt) reading frame overlapping out of phase the NH₂-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene; it **has been identified** as the animal equivalent of the recently discovered yeast H⁺-ATPase subunit 8 gene.

- **Better: subject and verb directly follow each other:**

The smallest of the URF's (URFA6L) **has been identified** as the animal equivalent of the recently discovered yeast H⁺-ATPase subunit 8 gene.

Subject-verb separation

- **Subject and verb are very much separated:**

Recently, however, immunoprecipitation **experiments** with antibodies to purified, rotenone-sensitive NADH-ubiquinone oxido-reductase [hereafter referred to as respiratory chain NADH dehydrogenase or complex I] from bovine heart, as well as enzyme fractionation studies, **have indicated** that **six human URF's** (that is, URF1, URF2, URF3, URF4, URF4L, and URF5, hereafter referred to as ND1, ND2, ND3, ND4, ND4L, and ND5) **encode** subunits of complex I.

- **Better: subject and verb directly follow each other:**

experiments [bla bla bla bla bla] **have indicated** that **six human URF's** [bla bla bla bla bla] **encode** subunits of complex I.

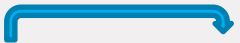
Emphasis

“Put in the topic position the old information that links backward; put in the stress position the new information you want the reader to emphasize.”

1. The backward-linking old information should appear in the topic position

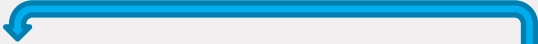
- Example

While doing this experiment, a problem with the calibration was observed. This problem will have to be corrected before we can move on.



- NOT

Before we can move on, we must correct this.



Emphasis - continued

2. The person, thing or concept whose story it is should appear in the topic position

- Example

The asphalt was heated, then cooled, over 100 times in rapid succession. Only minor cracking occurred.

- NOT

Our highly-skilled engineers tested the asphalt. They heated and cooled it, and they found only minor cracking occurred.

Emphasis - continued

3. The new, emphasis-worthy information should appear in the stress position

- Example

We are pleased to report that we have had **success**.

- NOT

This report of our **success**, which we are pleased to provide, follows.

Original text

The smallest of the URF's (URFA6L), a 207-nucleotide (nt) reading frame overlapping out of phase the NH₂-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene has been identified as the animal equivalent of the recently discovered yeast H⁺-ATPase subunit 8 gene. The functional significance of the other URF's has been, on the contrary, elusive. Recently, however, immunoprecipitation experiments with antibodies to purified, rotenone-sensitive NADH-ubiquinone oxido-reductase [hereafter referred to as respiratory chain NADH dehydrogenase or complex I] from bovine heart, as well as enzyme fractionation studies, have indicated that six human URF's (that is, URF1, URF2, URF3, URF4, URF4L, and URF5, hereafter referred to as ND1, ND2, ND3, ND4, ND4L, and ND5) encode subunits of complex I. This is a large complex that also contains many subunits synthesized in the cytoplasm.

Final text

The smallest of the URF's, URFA6L, has been identified as the animal equivalent of the recently discovered yeast H⁺-ATPase subunit 8 gene; but the functional significance of other URF's has been more elusive. Recently, however, several human URF's have been shown to encode subunits of rotenone-sensitive NADH-ubiquinone oxido-reductase. This is a large complex that also contains many subunits synthesized in the cytoplasm; it will be referred to hereafter as respiratory chain NADH dehydrogenase or complex I. Six subunits of Complex I were shown by enzyme fractionation studies and immunoprecipitation experiments to be encoded by six human URF's (URF1, URF2, URF3, URF4, URF4L, and URF5); these URF's will be referred to subsequently as ND1, ND2, ND3, ND4, ND4L and ND5.

Bonus: this is how ChatGPT made it easier to read

The smallest URF in URFA6L is a 207-nucleotide (nt) reading frame. It overlaps with the NH₂-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene. This URF has been identified as the animal equivalent of the yeast H⁺-ATPase subunit 8 gene. While the roles of other URFs have remained unclear, recent experiments using antibodies to purified, rotenone-sensitive NADH-ubiquinone oxido-reductase (also known as respiratory chain NADH dehydrogenase or complex I) from bovine heart have indicated something interesting. Enzyme fractionation studies have also supported this. They suggest that six human URFs (URF1, URF2, URF3, URF4, URF4L, and URF5) are actually encoding subunits of complex I. This complex is large and contains various subunits that are produced in the cytoplasm.

Look for logical gaps

- Explanations should be step-by-step
- You should not omit a step, assuming the reader can supply specialized knowledge
- Treat descriptions of processes like mathematical proofs; supply all steps

Topic action point-to-stress

Subject verb object

Problem solution benefit

Old transform into new

$A \rightarrow B.$

$B \rightarrow C.$

$C \rightarrow D.$

Summary

- Tell your story three times:
 1. Title
 2. Abstract
 3. Paper
- Place “old information” in the topic position for
 1. Linkage backward
 2. Creating context forward
- Follow the grammatical subject ASAP with its verb
- Finish off with “new information” in the stress position

Reporting CFD simulations

See Chapter 10.7 in the book by Versteeg and Malalasekera.

- **Inputs**

general description, code, geometry, boundary conditions, initial conditions, fluid/particle properties, grid information, solution algorithm, iterative convergence criteria, etc.

- **Results interpretation**

velocity vector plots, streamlines/trajectories, contours, profile plots, grid display, etc.
verification study, validation study, etc.